UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: COMPUTER NETWORKS

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SECTION : A

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LAB # 3

Lab 3 – Capture and Analyze Application Layer Traffic Using Wireshark

Introduction

In this lab, you will capture some application layer protocols traffic particularly focusing on HTTP and HTTPS protocols.

Task 1

- Launch Wireshark and start capturing traffic.
- Stop the capture and fill out the following tables.
- Write below the list of application layer protocols that were captured.

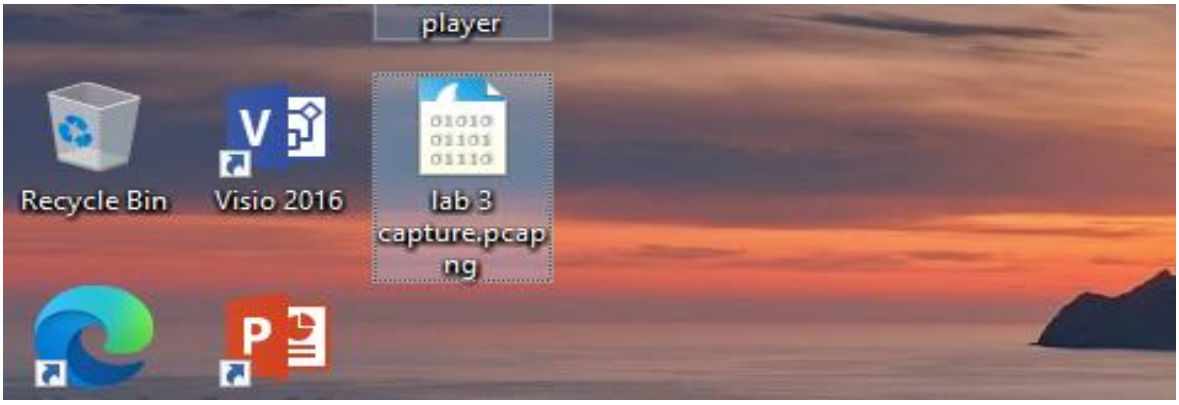
Time Source Destination Protocol

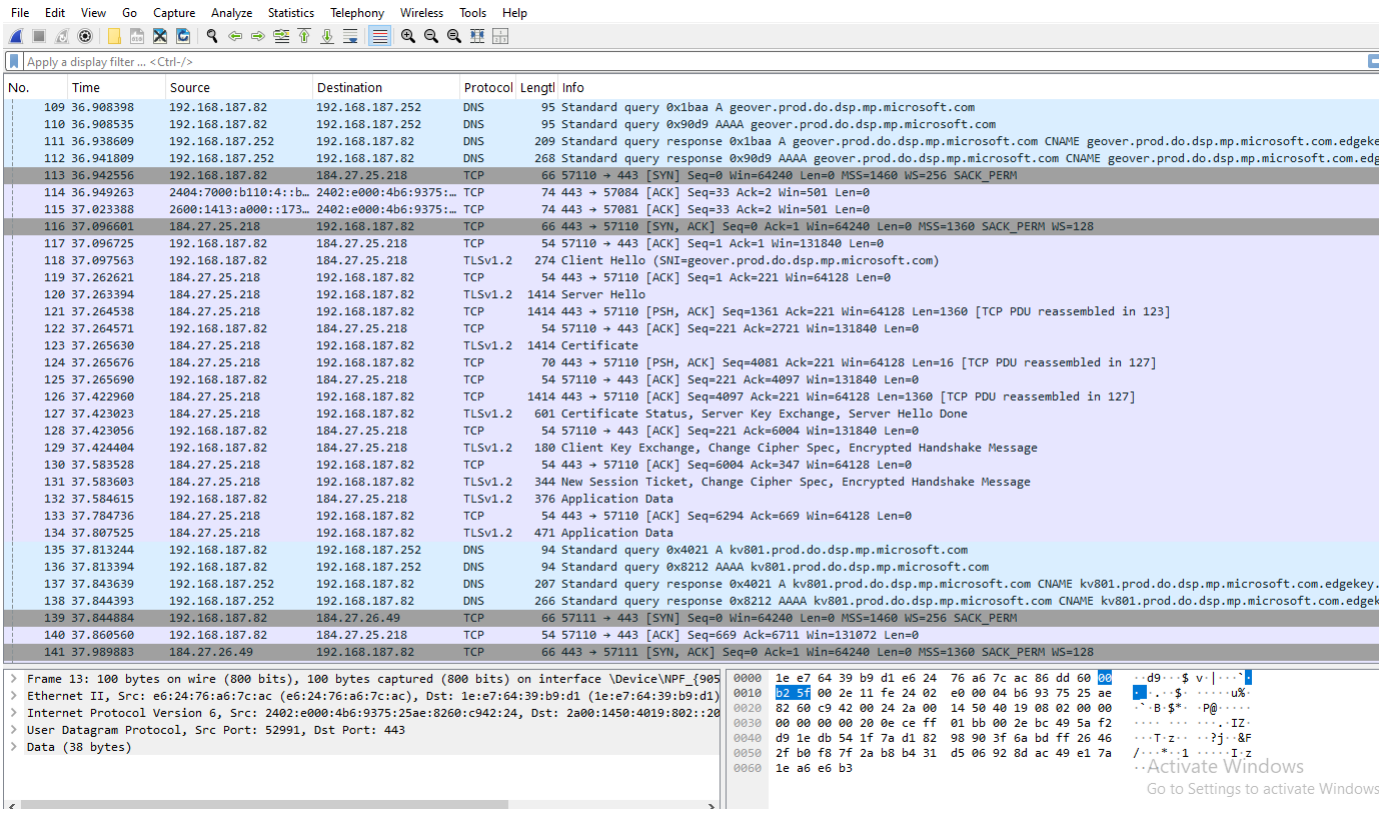
In this case, there are two application layer protocols (UDP and TCP) found in the capture.

No.	Time	Source	Destination	Protocol	Time
1	0.000	2402:e000:4b6:9375:25ae:8260:c942:24	2a00:1450:4019:802::200e	UDP	1
2	1.105	2a00:1450:4019:802::200e	2402:e000:4b6:9375:25ae:8260:c942:24	TCP	2

Lab Task 2

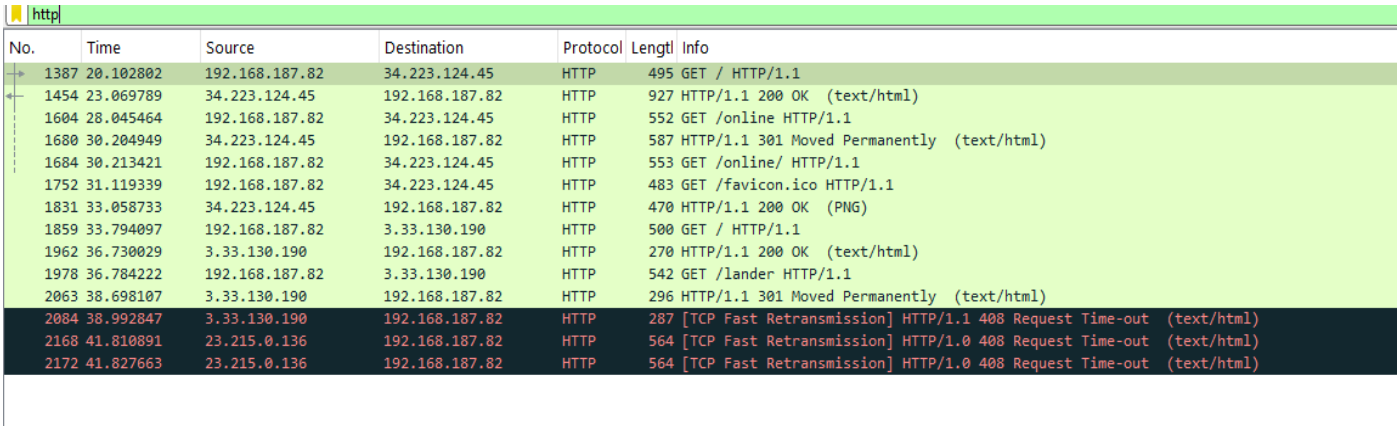
- Save the captured traffic by going to Files and then Save.
- Paste a screenshot of the Wireshark window showing the captured traffic.





Lab Task 3

- Start capture on Wireshark
- Open a website that uses HTTP (not HTTPS) for example, http://assami.kp.gov.pk/
- Stop the capture.
- Filter the captured traffic based on HTTP.
- Paste a screenshot of the filtered traffic below.
- Paste screenshots of all HTTP request messages.
- Paste screenshots of all HTTP response messages.
- From captured HTTP traffic identify all those packets in which images were received from the server.
    - Select one such packet and save the image to your computer by using 'Export Packet Bytes' option from the right click menu.



http request and response messages

request with the get

response with the ok

).	Time	Source	Destination	Protocol	Lengtl	Info
1387	20.102802	192.168.187.82	34.223.124.45	HTTP	495	GET / HTTP/1.1
1454	23.069789	34.223.124.45	192.168.187.82	HTTP	927	HTTP/1.1 200 OK (text/html)
1604	28.045464	192.168.187.82	34.223.124.45	HTTP	552	GET /online HTTP/1.1
1680	30.204949	34.223.124.45	192.168.187.82	HTTP	587	HTTP/1.1 301 Moved Permanently (text/html)
1684	30.213421	192.168.187.82	34.223.124.45	HTTP	553	GET /online/ HTTP/1.1
1752	31.119339	192.168.187.82	34.223.124.45	HTTP	483	GET /favicon.ico HTTP/1.1
1831	33.058733	34.223.124.45	192.168.187.82	HTTP	470	HTTP/1.1 200 OK (PNG)
1859	33.794097	192.168.187.82	3.33.130.190	HTTP	500	GET / HTTP/1.1
1962	36.730029	3.33.130.190	192.168.187.82	HTTP	270	HTTP/1.1 200 OK (text/html)
1978	36.784222	192.168.187.82	3.33.130.190	HTTP	542	GET /lander HTTP/1.1
2063	38.698107	3.33.130.190	192.168.187.82	HTTP	296	HTTP/1.1 301 Moved Permanently (text/html)

captured HTTP traffic identify all those packets in which images were received from the server.

1752	31.119339	192.168.187.82	34.223.124.45	HTTP	483	GET /favicon.ico HTTP/1.1
1831	33.058733	34.223.124.45	192.168.187.82	HTTP	470	HTTP/1.1 200 OK (PNG)
1859	33.794097	192.168.187.82	3.33.130.190	HTTP	500	GET / HTTP/1.1

THE END